

Session 22: Discover & Define

You've spent 21 sessions learning how AI works. Now it's time to put it all together and produce something you can show, share, and build on.

Over the next **four sessions**, you'll write a research paper investigating how AI is being applied in a domain you choose. The paper uses a structure and workflow from real researchers. By the end, you'll have produced something you can show, share, and build on.



The Capstone Topic

The capstone topic is fixed at the umbrella level: **Applications of AI in Various Domains**. Within that umbrella, you choose your specific focus.

Healthcare

Diagnostics, patient care, allergy management

Sports

Training apps, performance analysis, coaching

Music

Emotion-aware generation, music education

Education

Math practice, tutoring, student dependency

Wildlife

Citizen science, species identification, conservation

Accessibility

Communication tools, assistive technology

 You're not investigating all of AI. You're investigating how AI is being applied in **one specific area you care about**.

What This Paper Is

A Real Investigation

A research paper investigating how AI is being applied in a real-world domain using the same structure real researchers use.

Literature + Hands-On

You'll read what's been written about AI in your domain, then do a small hands-on exploration with a tool that fits your investigation.

A Synthesis

The paper synthesizes what the literature says with what your hands-on exploration reveals - two threads woven together.



What This Paper Is *Not*

Not an Engineering Paper

You won't be measuring statistical performance or reporting accuracy numbers as findings.

Not a Technology Claim

You're not claiming to have produced new technology. The hands-on exploration is a **learning activity**.

Not Exploration-Centered

Your observations from the exploration become part of your paper, but they're not the paper's main contribution.

Examples of Research Questions

These are the kinds of questions students investigate in this capstone. Notice the pattern: each names a **domain**, a **specific application**, and **what's being investigated**.

Healthcare

How is AI being applied to support food allergy management for children, and what bias and safety considerations come with it?

Sports

How is AI being used in youth basketball training apps, and how effectively does it support skill development?

Music

How is AI being applied to emotion-aware music generation, and what does this enable for music education?

Education

How is AI being applied to early elementary math practice, and what concerns arise about student dependency?

Wildlife

How is AI being used in citizen science wildlife identification, and what are the implications for conservation efforts?

Today's Goals

By the end of this session, you'll have completed all of the following:

01

Choose Your Domain

Selected a domain and a specific focus within it

02

Sharpen Your Question

Refined your research question so it passes the Three-Part Test

03

Understand the Structure

Understood the paper structure you'll be writing toward

04

Find Sources

Found 5–8 candidate sources using the three-tool workflow

05

Sketch & Outline

Sketched your Abstract and outlined your Introduction

06

Commit to Exploration

Committed to a hands-on exploration plan for next session



The Personalization Moment

For the past 21 sessions, your mentor has watched what you engage with, which sessions lit you up, which tools you took to, which problems you cared about, which industry conversations caught your attention.

Based on those signals, your mentor has prepared **two or three specific paper angles tailored to you**. These aren't random. They're grounded in what your mentor has seen across 21 sessions of your work.

How to Choose Your Angle

Listen to all the angles your mentor offers. For each one, your mentor will explain:

The Domain & Focus

The specific domain and application area being proposed

Why It Matches You

Based on observations from Sessions 1–21 of your work

The Research Question

The kind of question this angle would produce

Existing Sources

What sources already exist on this topic

The Hands-On Exploration

What you'd do as your learning activity for Session 23

- ① Pick the one you **actually want to investigate**, not the one that sounds the smartest. You're going to spend four sessions on this - pick what holds your interest.

The Paper Structure

Your paper will have **eight sections** plus an AI Attribution Statement and References.

1 - Abstract	2 - Introduction
3 - Related Work	4 - Methodology
5 - Bias & Fairness	6 - Safety & Governance
7 - Results	8 - Conclusion & Future Work

Plus an **AI Attribution Statement** and **References** at the end.

Section 1 - Abstract

What It Is

The Abstract is **one paragraph** that summarizes the entire paper. It tells a reader:

- What the paper investigates
- How it was investigated
- What was found
- What it means

Your Task Today

Sketch a **placeholder version (~80 words)** today. The real Abstract gets written in Session 24, once you know your actual findings.

Section 2 - Introduction

The Hook

Opens with a compelling reason why your domain matters

Context

Gives background on what's already known about AI in your domain

Research Question

Ends with your sharp, specific research question

The Introduction sets up your paper. It opens with a hook that shows why your domain matters, gives context about what's already known, and ends with your research question. You'll outline this in bullets today and expand it to prose in homework.

Section 3 - Related Work

What It Is

Related Work synthesizes what's known about AI in your domain. It draws on your sources to show what other researchers, journalists, and developers have found.

When You Write It

You'll draft this in **homework after today**, once you've identified your 5-8 verified sources.

Key Skill

Finding the threads that run across multiple sources the agreements, the tensions, the gaps and weaving them into a coherent picture.

Section 4 - Methodology



Research Workflow

Document the tools you used to find and verify sources, and how you approached the literature.



Hands-On Exploration

Describe the tool you used, what you explored, and how you structured your firsthand investigation.



How You Put It Together

Explain how you combined literature findings with hands-on observations to build your investigation.

 You'll draft Methodology in **Session 23**, after completing your hands-on exploration.

Section 5 - Bias & Fairness Analysis

What It Examines

This section examines bias issues in AI applications across your domain - drawing on what the literature shows and on what you observed in your hands-on exploration.

Frameworks You'll Apply

You'll apply the **bias frameworks from Session 20** to analyze your domain. Think about:

- Who the AI was trained on
- Who it works well for and who it doesn't
- What the literature says about fairness in your domain

You'll draft this in **Session 24**.

Section 6 - Safety & Governance

What It Covers

This section examines safety and governance concerns in AI applications across your domain - risks, regulations, oversight, and accountability.

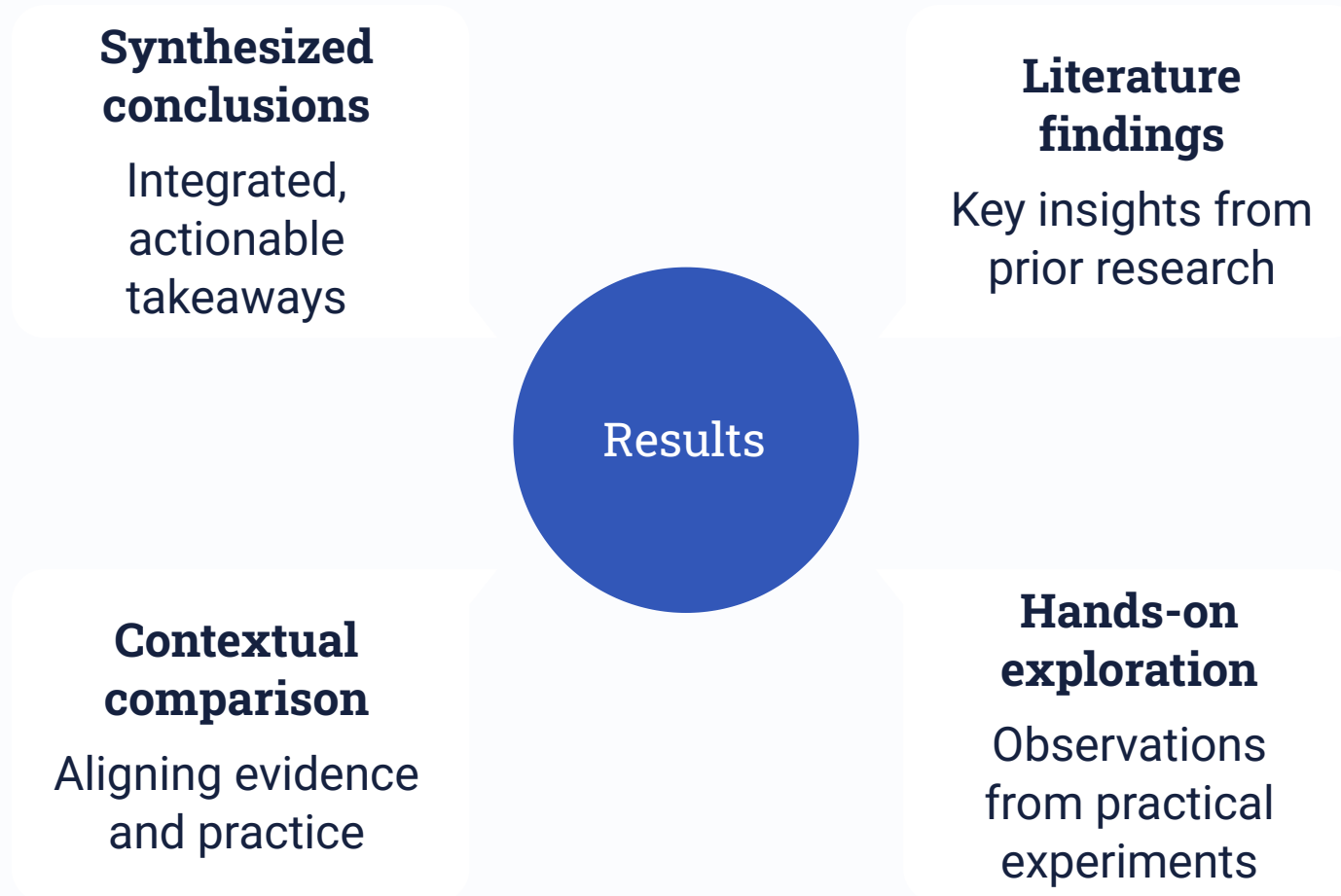
Frameworks You'll Apply

You'll draw on the **safety frameworks from Session 21** to structure your analysis of what safeguards exist and what gaps remain.

When You Write It

You'll draft this in **Session 24**, alongside the Bias & Fairness Analysis section.

Section 7 - Results



Section 8 - Conclusion & Future Work

Conclusion

Summarizes what your investigation showed about AI in your domain. What did you find? What does it mean? What should readers take away?

Future Work

Names what would come next questions you didn't answer, areas that need more investigation, directions a future researcher could take.

 You'll draft both parts in **Session 24**.

AI Attribution Statement

What It Is

A short statement disclosing which AI tools you used in your research and how you used them.

Why It Matters

This is a **normal practice in modern research** being transparent about AI assistance. Researchers and journalists are increasingly expected to disclose when and how AI tools contributed to their work. You'll write this in **Session 24**.

What Makes a Good Research Question

A **topic** is broad. A **research question** is sharp. A good research question passes three tests:



Specific

It names a domain, a specific application within it, and what's being investigated.



Original

You're investigating something - comparing, analyzing, examining - not summarizing one source.



Achievable

You can answer it in four sessions using literature research and a hands-on exploration.

Sharpening a Research Question

Too Broad	Sharpened
AI in Healthcare	How is AI being applied to support food allergy management for children, and what bias and safety considerations come with it?
AI in Education	How is AI being applied to early elementary math practice, and what concerns arise about student dependency?
AI in Music	How is AI being applied to emotion-aware music generation for educational settings?
AI in Sports	How is AI being used in youth basketball training apps, and how effectively does it support skill development for ages 10-14?

❗ Each sharpened question names a **domain**, a **specific application**, and a **clear focus** for investigation.

The Hands-On Exploration

Alongside your literature research, you'll do a small hands-on exploration using a tool that fits your domain. Tools you've used across Sessions 1-21 are a good starting point:

Teachable Machine

ChatGPT

NotebookLM

Lovable

ElevenLabs

Veo 3

Pictory

Gemini

Perplexity

- ① You can also use other tools - anything that helps you experience how AI works in your domain. Pick the tool that **fits what you're investigating**, not just the tool you happen to know best.

Examples of Hands-On Explorations

Domain	Possible Hands-On Exploration
Food allergy management	Set up a small Teachable Machine classifier with peanut and non-peanut food images. Observe what works, where it struggles, what surprises you.
Music education	Use ElevenLabs to generate the same sentence in different emotional tones. Note how the AI handles emotional expression.
Early math learning	Use ChatGPT to generate practice problems at different grade levels. Observe quality, difficulty, and any errors.
Wildlife identification	Set up a Teachable Machine classifier for two local species, or try Merlin or iNaturalist and analyze how it works.
Pediatric health communication	Use Lovable to mock up an app interface that explains medication labels, or explore AI for accessibility.

Finding Sources

You need to find **5-8 credible sources** that discuss AI in your domain. A common workflow uses three tools in order:

Perplexity

Discover candidates - collect 8-12 sources with citations

Gemini

Compare and filter - narrow down to the most relevant

NotebookLM

Verify content - confirm each source is real and accessible

ⓘ This is a **recommended workflow**, not the only valid one. What matters is that your sources are real, credible, and actually say what you think they say.

Perplexity

What It Does

Perplexity is a search tool that returns answers with **citations to real sources**. Every claim links to a source you can click and verify. It's well suited to discovering candidate sources by searching your research question.

Today's Goal

Collect **8–12 candidate sources** from Perplexity searches. If you're using a different discovery tool, your goal is the same - collect 8-12 candidates with verifiable citations.

The Source Credibility Checklist

For each candidate source, ask these five questions. Pass threshold: **4 of 5**. Drop sources that fail more than one check.

Question	Pass Criterion
Is the author named?	Author has credentials or affiliation listed
Is the publisher credible?	Peer-reviewed journal, university, government, established news, or reputable publication
Is the publication date within 3 years?	AI moves fast; older sources may be outdated
Can you find the original source?	If cited from a summary, the original exists and is accessible
Will NotebookLM load the content?	The source is accessible and parseable

Sources to Avoid

Some kinds of sources don't pass the credibility check. Avoid these:

✗ Random Blogs

Without identified authors or institutional affiliation

✗ Wikipedia

Use it for orientation only cite the original sources Wikipedia points to

✗ AI-Generated Articles

Without independent verification of claims and sources

✗ Predatory Journals

Pay-to-publish without rigorous peer review

✗ Social Media Posts

Not a credible primary source for research

✗ Marketing Content

Advertising or promotional material from companies

Reminder: AI Can Hallucinate Sources

Verify every source actually exists before adding it to your list. Don't trust a citation just because it looks real.

- ① AI tools can generate citations to papers that don't exist. This is a known failure mode of large language models.

Today's Workflow

Now you'll work through the session with your mentor coaching throughout:



Choose Your Angle

Select from your mentor's options



Sharpen Your Question

Refine with mentor coaching until it passes the Three-Part Test



Sketch Your Abstract

~80 words placeholder version



Outline Your Introduction

Bullet points - prose comes in homework



Run Source Discovery

8–12 candidates → apply checklist
→ cut to 5–8 → verify



Commit to Exploration Plan

Name your tool and what you'll do in Session 23

Homework

Homework takes about **45–60 minutes** total. Complete all tasks before Session 23.

Verify each source. Write a one-sentence summary of each - in your own words.

Draft your **Related Work** section. Synthesize across sources - don't summarize them one by one.

Expand your **Introduction** outline into prose (100–200 words). Reference at least two sources.

- ⊗ Use direct paraphrasing in Related Work. **Don't copy phrasing from sources.** If you can't say it in your own words, you don't understand it well enough yet.

What's Next - Session 23

Bring to Session 23

- Your source list with one-sentence summaries
- Your Related Work draft
- Your Introduction draft
- Your final hands-on exploration plan with any adjustments

What Happens in Session 23

Research & Build. You'll do the deep literature read and your hands-on exploration. You'll draft Methodology and start Results.

By end of next session, you'll have **four paper sections drafted**.